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Performance of GPM-IMERG satellite precipitation for rainfall-runoff modeling in Indonesia

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ABSTRACT

Accurate discharge predictions are essential for effective water resource management, including irrigation, drinking water supply, hydropower, reservoir management, environmental flow, and flood assessment. Satellite precipitation data, such as GPM-IMERG, offers a viable solution for estimating river discharge, especially in ungauged or sparsely gauged watersheds. This study evaluates the use of GPM-IMERG data in the HEC-HMS hydrological model to predict discharge in the Mayang Watershed, Jember Regency, Indonesia. The study employs the Arithmetic, Polygon Thiessen, and Isohyet methods to analyze rainfall distribution. Results indicate that the Polygon Thiessen method, when paired with GPM-IMERG data, provides more reliable precipitation estimates due to its consideration of rain gauge proximity. Calibration and validation confirm the superiority of GPM-IMERG data over traditional observational data, particularly under high-flow conditions. Despite some limitations in detecting low-intensity precipitation, GPM-IMERG outperforms other satellite products like TRMM in terms of accuracy. Sensitivity analysis highlights the significant influence of parameters such as curve Number and time of concentration on model outcomes. Further research should explore emerging rainfall data products to improve hydrological modeling accuracy.

Key words: GPM-IMERG satellite data, HEC-HMS, Mayang, rainfall-runoff modeling, rain gauges, Thiessen polygons

HIGHLIGHTS

- GPM-IMERG satellite data combined with HEC-HMS modeling more accurately describe area rainfall without observation gage cover area.
- The Polygon Thiessen method enhances rainfall estimation accuracy, leading to higher NSE values.
- GPM-IMERG satellite data simulation is better than TRMM satellite data based on NSE value.
- The most sensitive parameters affecting the manual calibration are lag time and curve number.

1. INTRODUCTION

Accurate discharge predictions are critically needed for the management of water resources, including irrigation, drinking water supply, hydropower, reservoir management, environmental flow, flood assessment, and other related activities. Analyzing flood hydrographs, particularly in unmeasured basins, can be facilitated by utilizing satellite precipitation data or hydrological models to estimate river discharge. Prediction of runoff can be achieved by employing rainfall-runoff modeling. Hydrologic Engineering Center - Hydrologic Modeling System (HEC-HMS) is a common rain flow modeling tool to used worldwide. This program is easy to use because hydrological modeling can be done with rainfall and discharge data as input and can perform various roles, one of which is to estimate the maximum flow rate of a watershed (Febriyanto *et al.* 2018). Based on the previous study, HEC-HMS generally produced acceptable calibration and validation results, performing slightly better than Soil and Water Assessment Tool (SWAT) during the calibration phase for the Manafwa catchment, which statistical tests such as the *t*-test confirmed (Sempewo *et al.* 2023). However, precise runoff modeling encounters challenges due to the limited availability of rainfall data, as not all watersheds include rain gauges that are uniformly distributed across the whole watershed namely the Mayang Watershed.

A potential way to address the absence of field rainfall data is to utilize satellite precipitation data to substitute for missing or incomplete rain gauge data. Satellite precipitation data is capable of capturing rainfall information by

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remotely measuring or recording rainfall, a task that is challenging for rain gauges deployed in the field. Satellite rain data products provide the benefits of being readily accessible, providing real-time information, and offering extensive coverage in terms of both time and space (Yeditha *et al.* 2022). The availability of satellite data enables the acquisition of rainfall information at any location and time. The Global Precipitation Measurement – Integrated Multi-Satellite Retrieval for GPM (GPM-IMERG) is one of the satellites that collects rainfall data globally. GPM-IMERG provided adequate spatiotemporal coverage and was capable of monitoring the precipitation rate in near real time, utilizing passive microwave and infrared satellite observations to provide rain rate estimates every 30 min in $0.1° \times 0.1°$ fields (Giannaros *et al.* 2022). A study by Saouabe *et al.* (2022), found that while the GPM-IMERG data slightly underestimated the amount of rainfall, it had a good correlation with ground observations and showed smaller mean error values, indicating better performance in certain metrics. Based on previous study in Bedadung Watershed, Jember Regency, Indonesia, the uses of GPM-IMERG outperform the Tropical Rainfall Measuring Mission (TRMM) for various hydrological purposes, suggesting that it provides a more accurate representation of rainfall data (Hidayah *et al.* 2021). Furthermore, based on the study by Ahmed *et al.* (2020), the GPM-IMERG is considered a suitable replacement for TRMM 3B42 for rainfall-runoff modeling in ungauged watersheds because of higher correlation coefficients, better Nash–Sutcliffe efficiency (NSE) values, and lower percent biases, offering better accuracy and reliability in Chenab River, Pakistan.

The main technologies used in the collection of rainfall data by the GPM and TRMM satellites involve advanced instruments such as the precipitation radar and dual-frequency precipitation radar, which emit microwave signals to measure the intensity and vertical distribution of rainfall (Ryu *et al.* 2021; Gorooh *et al.* 2022). Additionally, the TRMM Microwave Imager and GPM Microwave Imager detect natural microwave radiation to estimate rainfall intensity, while infrared sensors measure cloud temperature to identify potential precipitation (Sekaranom & Masunaga 2019; Milani *et al.* 2022). These satellites regularly orbit the Earth, collecting data from various regions globally, and the data collected is processed and validated with ground-based observations to ensure accuracy (Seela *et al.* 2024). This process enables continuous monitoring of rainfall worldwide, including in areas that are difficult to reach by conventional measurement methods (Ryu *et al.* 2021; Fusco *et al.* 2022; Li *et al.* 2023).

The Mayang Watershed is characterized by a small size and an unequal distribution of rain gauges, as well as variations in topography. Consequently, it is necessary to evaluate the average amount of precipitation across a certain area. Therefore, this study will examine the distribution of rainfall using the Arithmetic, Polygon Thiessen, and Isohyet methodologies due to the small size and highland location of the studied area. This study aims to predict discharge data based on satellite precipitation data GPM-IMERG using the HEC-HMS program in the Mayang Watershed, Jember Regency. Talang station serves as the designated outflow point for the calibration and model validation of the Mayang Watershed, namely for observation discharge control. The sensitivity test is used to identify the most influential parameters affecting the model outputs. The model results will be reassessed with various temporal variations of rainfall data and sensitivity analyses on the parameters will be conducted.

2. METHODS

2.1. Study area

The study is conducted in the Mayang Watershed, located in Jember Regency, East Java province. The watershed has an area of 537.4631 square kilometers. The flooding in Wonosari Village, Tempurejo Sub-district in early 2021 was worsened by the accumulation of sediment in the Mayang watershed. The geographical location of the place is defined by the coordinates 7.403509°–8.498642° South Latitude and 113.285950°–114.391713° East Longitude. The Mayang watershed has facilities with 12 rain stations, namely Silo, Sumber Jati, Ledokombo, Ajung, Suren, Pakusari, Jatian, Seputih, Talang, Wirolegi Karang Kedawung, and Jenggawah, as shown in Figure 1. The Talang station serves as the outlet point for the watershed.

The study area encompasses six distinct land use categories: residential and public areas, forests, gardens, fields, rice fields, and shrubs as the detailed shown in Table 1. The soil type is utilized in the classification of soil type and land use to ascertain the value of the curve number (CN). Furthermore, there are two types of soil in this area which are andosol and Mediterranean.

2.2. Data preparation

This study includes using hydrological and geographical data. The hydrological data includes the daily rainfall data and the daily discharge observed data (2012–2021) was obtained from Binamarga PU and SDA Jember

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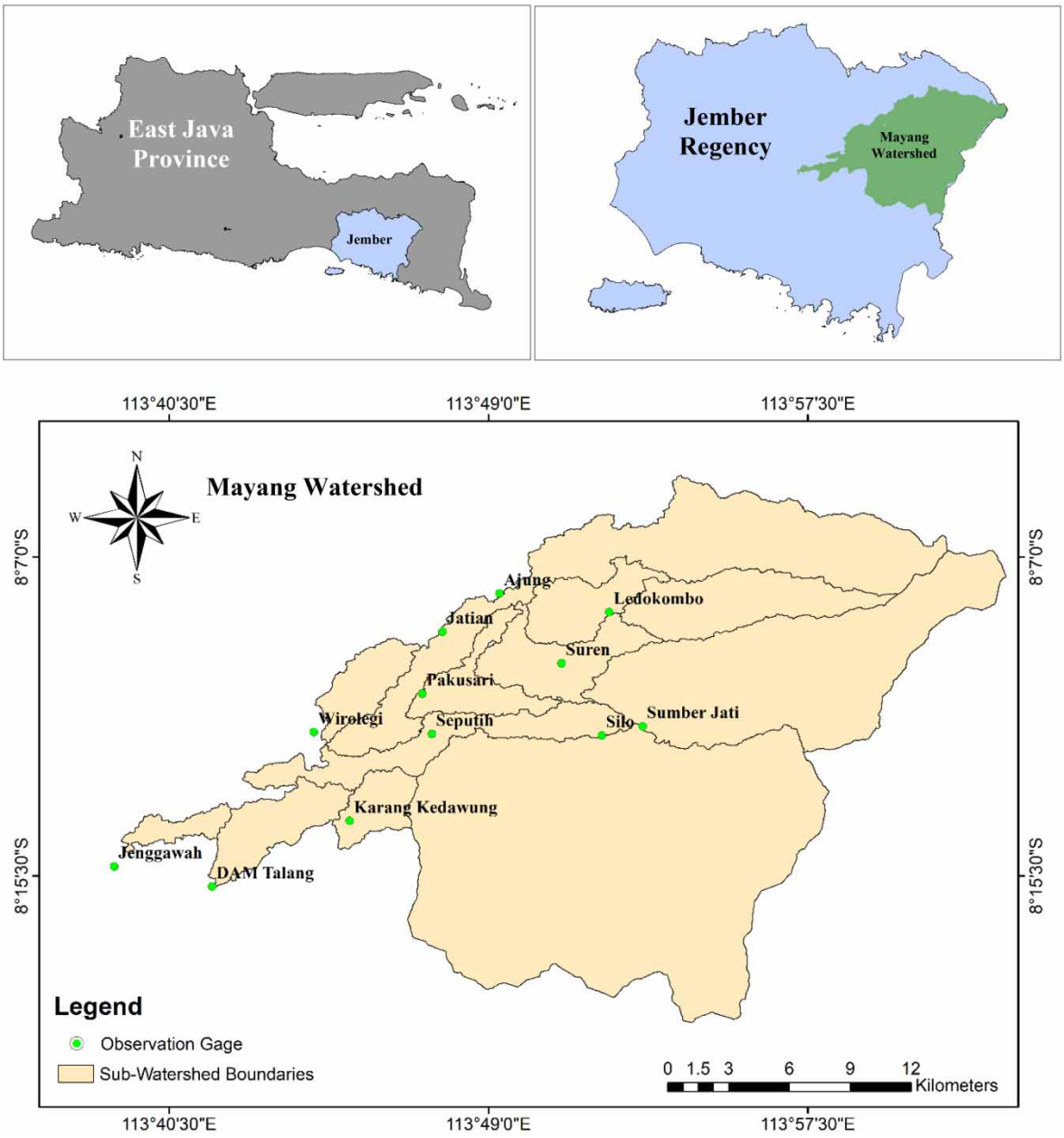


Figure 1 | Study area.

Table 1 | Distribution of land use

No.	Land use	Areas (Km^2)	Percentage (%)
1	Residential	50.417	9.380,550,962
2	Forest	129.695	24.130,958,94
3	Garden	179.576	33.411,782,13
4	Field	60.5781	11.271,117,96
5	Rice field	103.328	19.225,133,78
6	Shrub	13.869	2.580,456,221

Regency, along with the GPM-IMERG satellite rain data spanning a decade (2012–2021) was acquired from the website https://giovanni.gsfc.nasa.gov/giovanni/. The available observed data with good and consistent quality only covers a period of 10 years, while longer records have poor data quality. Moreover, the geographical

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data, including the digital elevation model data, were sourced from the website https://tanahair.indonesia.go.id/demnas/#/demnas, and the land use map (2017) was obtained from the website https://tanahair.indonesia.go.id/portal-web/download/perwilayah.

2.3. Methodology

The study methodology was conducted in multiple stages, as shown in Figure 2, with the objective to obtain good modeling results. This stage involves rain and watershed analysis, along with HEC-HMS rainfall-runoff simulation.

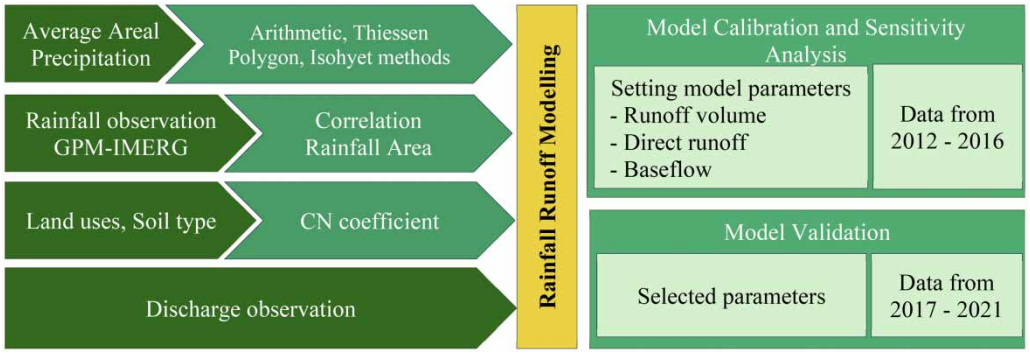


Figure 2 | Research methodology.

2.3.1. Rainfall-runoff data processing

During the data processing stage, a rainfall correlation analysis was conducted, followed by calculating the area average utilizing both observational and satellite rainfall data from GPM-IMERG.

2.3.2. Correlation of observational rainfall data and GPM-IMERG satellite precipitation data

It is important to acknowledge potential limitations in the accuracy or spatial resolution of the GPM-IMERG precipitation data, as these factors could impact the model results. For instance, lower spatial resolution may lead to discrepancies in localized rainfall patterns, potentially affecting the accuracy of the simulated discharge in specific areas within the watershed (Xiaojun *et al.* 2021). Additionally, inaccuracies in the satellite rainfall data, such as errors in rainfall estimation algorithms or limitations in ground validation data, could introduce uncertainties into the model outputs (Wiwoho *et al.* 2021). Therefore, this study is also followed by correlation, which aims to show the value of the relationship between the observed value and the simulated value. Correlation is expressed by a coefficient that shows the related linear relationship between two changes. The correlation Equation R is shown in Equation (1) (Sharma *et al.* 2020). Rain data correlation using observational rain data and the GPM-IMERG satellite from 1 January 2012 to 31 December 2021 was carried out by software Excel with daily, 10 daily, and monthly correlation.

$$R = \frac{n\sum xy - \sum x \sum y}{\sqrt{n\sum x^2 - (\sum x)^2}\sqrt{n\sum y^2 - (\sum y)^2}} \quad (1)$$

where X is observation rain data, and Y is satellite rain data.

If the correlation coefficient value shows a larger value between the two, it means that the relationship between the two is getting stronger. In other words, the satellite data estimation pattern is getting closer to the actual data pattern at the observation rain station (Ng *et al.* 2023). The following justification can be drawn from the correlation coefficient value: 0 (no correlation between two variables), 0–0.25 (very weak correlation), 0.25–0.5 (moderate correlation), 0.5–0.75 (strong correlation), 0.75–0.99 (very strong correlation), and 1 (perfect correlation) (Octafiani Hartawan *et al.* 2021).

2.3.3. Mean areal precipitation

The watershed encompasses multiple rainfall stations, necessitating the computation of average rainfall to gauge precipitation across the area. Three methods used to calculate the area's average rainfall for observational and

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satellite rain data in this study are the algebraic, Thiesen, and Isohyet methods because the watershed area is small. The Arithmetic method offers simplicity by summing rainfall measurements from all gauges and dividing by the total count, but it overlooks spatial variability, assuming uniform distribution (Ribeiro *et al.* 2021; Zakaria *et al.* 2023). In contrast, the Thiessen polygon method partitions the watershed into polygons around each gauge, weighting rainfall data based on proximity, thus addressing spatial variability (Ribeiro *et al.* 2021). However, it assumes uniform rainfall within polygons and may not adequately represent distant areas (Smith *et al.* 2023). Meanwhile, the Isohyet method delineates lines of equal rainfall intensity across the watershed, considering both spatial and topographic variability but demanding more intricate calculations (Workneh *et al.* 2024). The Arithmetic, Thiessen, and Isohyet methods, as shown in the following equations (Ribeiro *et al.* 2021).

Arithmetic method:

$$P = \frac{P1 + P2 + P3 + \cdots + Pn}{n} = \frac{\sum_{1=1}^{n} P_i}{n} \tag{2}$$

Thiessen Polygon method:

$$P = \frac{(A1P1) + (A2P2) + \cdots + (A_iP_i)}{(A1 + \mathrm{A2} + \cdots An)} = \sum_{i=1}^{n} \frac{P_i * A_i}{A} \tag{3}$$

Isohyet method:

$$P = \frac{A1\dfrac{(P1 + P2)}{2} + \cdots An\dfrac{(Pn + Pn)}{2}}{(A1 \ldots An - 1)} = \sum_{i=1}^{n} \frac{P_i * C_i}{A} \tag{4}$$

where P is rainfall rate (mm), P_i is the daily precipitation at i location, n is the number of rain gauge, C_i is the average rainfall between the isohyet, and A is area (km^2). Additionally, this study will compare the best mean areal precipitation method results, integrated with GPM-IMERG, with those of TRMM to evaluate their accuracy and performance in predicting discharge data.

2.3.4. HEC-HMS rainfall-runoff modeling

HEC-HMS is a program that aims to produce a simulated discharge value based on the value of the rainfall data by considering the observed discharge value. In this study, following spatial data analysis using the Arc-GIS tool, hydrological modeling can be conducted to simulate discharge. The quasi-distributed is used which divides the watershed into multiple sub-basins, allowing for spatial variability in parameters and inputs. This process involves several steps. The basin model is the initial step in establishing a project in the HEC-HMS program. It primarily involves creating important hydrological components such as subbasins, reaches, junctions, and sinks. Subsequently, Terrain data are produced by integrating the Mayang watershed boundary map obtained from Arc-GIS delineation in .tif format. This involves navigating the Geographic Information System (GIS) menu and doing actions such as preprocess sink, preprocess drainage, and delineate element. These tasks lead to the creation of subbasins, flow direction, flow accumulation, reaches, and junctions. In the meteorologic model step, precipitation calculations are chosen. This study specifically uses the specified hyetograph method, which entails using rainfall values that correspond to a 5-year return period for each sub-basin. The control specification establishes the timeframe for the execution of the process, using rainfall data from 1 January 2012, to 31 December 2021, with a daily frequency for both rainfall and discharge. Time series data utilizes different forms of data as inputs for modeling, such as recorded rainfall, GPM-IMERG precipitation data, and recorded daily discharge throughout a 5-year period from 1 January 2012, to 31 December 2016. The rainfall data are obtained by calculating the average amount of rainfall from 12 stations located within the Mayang watershed, measured in millimeters. On the other hand, the discharge data refers to the rate at which water flows into the Talang outlet, measured in cubic meters per second. Finally, the initial condition entails the computation of parameter values prior to the execution of the model. These values are derived through manual calculations using methods supplied in the HEC-HMS Technical Reference Manual. The resulting values are then inserted into the selected runoff calculation technique or model. These methods and parameters selected for HEC-HMS modeling are detailed in Table 2.

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Table 2 | HEC-HMS parameters modeling

Model	Method	Parameters
Canopy	Simple canopy	Initial storage (IS) (%) Max storage (MS) (mm)
Surface	Simple surface	Initial storage (IS) (%) Max storage (MS) (mm)
Runoff volume	SCS CN	Initial abstraction (IA) (mm) Curve number (CN)
Direct runoff	SCS UH	Time of concentration (TC) (h) Storage coefficient (SC) (h)
Baseflow	Constant monthly	Base flow per month (m^3/s)

This study used the loss method, transform method, and baseflow method to do hydrological modeling. The calculation for the loss method parameter is a method to calculate runoff volume. In this study, the SCS CN method is used. The parameters used in the Soil Conservation Service-Curve Number (SCS-CN) method are the CN, initial abstraction (I_a), and imperviousness. In this study, calculations of volume *runoff* using SCS CN. This method is widely used because it is simple, predictable, and stable by following the formula, as shown in the following equation (Kastridis *et al.* 2020).

$$P_e = \frac{(P - I_a)^2}{P - I_a + S} \tag{5}$$

The relationship between the maximum storage capacity value and the watershed characteristic value can be seen by the CN value can be seen in the following equation (Lee *et al.* 2023):

$$Ia = 0.2\,x\,S \tag{6}$$

$$P_e = \frac{(P - 0.2S)^2}{P + 0.8S} \tag{7}$$

$$S = \frac{25,400 - 254.\ \mathrm{CN}}{\mathrm{CN}}\ (\mathrm{mm}) \tag{8}$$

A watershed that has various kinds of soil and land use requires a composite CN value, as shown in the following equation (Ranjan & Singh 2022):

$$\mathrm{CN}_{\mathrm{composite}} = \frac{\sum A_i\ \mathrm{CN}_i}{\sum A_i} \tag{9}$$

where P_e is effective precipitation or runoff (mm), P is total precipitation (mm), I_a is a initial abstraction (mm), S is potential maximum retention after runoff begins (mm), and CN is a dimensionless parameter used in the SCS CN method to represent the combined effect of soil type, land use, and hydrological condition of a watershed on runoff potential. The calculation for the transform method serves the same purpose as the loss method, which is to find the direct runoff volume. The method used in this study is the SCS unit hydrograph. The parameter used in the SCS unit hydrograph method is the lag time (t_{lag}) (Niyazi *et al.* 2022), with the selected graph type being standard. This parameter is based on the catchment area or rain station. The formula used to find the value of making time is shown in the following equation (Ben Khélifa & Mosbahi 2022).

$$t_{lag} = 0.6\ \times t_c \tag{10}$$

$$t_c = \frac{100\ L^{0.8}\left(\frac{1,000}{\mathrm{CN}} - 9\right)^{0.7}}{1,900\ S^{0.5}} \tag{11}$$

where t_{lag} is lag time (h), t_c is the time of concentration (h), L is the length of the longest flow path (m), and S is potential maximum retention after runoff begins (mm). Baseflow is the flow of rainfall retained in the soil. The

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baseflow method used in this study employs the constant monthly method. The required parameter is the baseflow value for each month by dividing the area per subbasin by the total subbasin area and then multiplying by the minimum monthly flow (Fadhilla & Lasminto 2021).

2.3.5. Sensitivity analysis and model performance

Sensitivity analysis is used to identify the parameters that significantly influence model responses and to analyze their interactions, range of choices, and spatial variability to enhance model outcomes (Devak & Dhanya 2017). This process involves determining the impact of various parameters on the calibration process and their interactions with other parameters (Li *et al.* 2023). Parameter sensitivity is measured by the magnitude of the increase or decrease in the values of the differences, which is the greater the magnitude of the difference, the higher the sensitivity of the parameter.

Calibration is performed to establish the initial parameters (initial conditions) utilized for modeling that align with the observed hydrograph. If the given conditions are not met, adjustments are performed. However, if the conditions are met, the process can proceed to the validation stage. The calibration period for this study spans from 1 January 2012 to 31 December 2016. Validation is conducted to see whether the parameters align with the calibration stage. The validation period is from 1 January 2017 to 31 December 2021. The calibration and validation using the NSE, Root Mean Square Error (RMSE), and percent bias method, as shown in the following equations (Wang *et al.* 2023; Xu *et al.* 2023).

$$\mathrm{NSE} = 1 - \frac{\sum i(Q\mathrm{sim} - Q\mathrm{obs})^2}{\sum i(Q\mathrm{obs} - \overline{Q\mathrm{obs}})^2} \tag{12}$$

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{n} (Q\mathrm{sim}_i - Q\mathrm{obs}_i')^2}{n}} \tag{13}$$

$$\mathrm{PBIAS} = \left[\frac{\sum_{t=1}^{T} (Q\mathrm{sim}_o^t - Q\mathrm{sim}_m^t) \times 100}{\sum_{t=1}^{T} Q\mathrm{sim}_o^t} \right] \tag{14}$$

where Qsim is the modeled discharge (m^3dt), Qobs is the observed discharge (m^3dt), and $\overline{Q\mathrm{obs}}$ is the average observed discharge (m^3dt). NSE with a value that ranges from 0 to 1 with the closer value being the better value. The accuracy value of NSE can be categorized, as shown in Table 3 (Lee *et al.* 2021).

Table 3 | NSE value accuracy category

Category	NSE
Very good	$0.75 < \mathrm{NSE} < 1.00$
Good	$0.65 < \mathrm{NSE} < 0.75$
Enough	$0.5 < \mathrm{NSE} < 0.65$
Bad	$\mathrm{NSE} < 0.5$

3. RESULTS

3.1. Average areal precipitation

In this study, the Arithmetic, Thiessen Polygon, and Isohyet techniques are approached for calculating rainfall distribution. Each method takes a unique approach to determining rainfall distribution across a research area. Figure 3(a) represents the Arithmetic method. This method implies that rainfall is distributed uniformly across the whole study region, regardless of any geographical or environmental factors that may influence precipitation patterns. While this method is simple and easy to use, it does not take into consideration the regional variability of rainfall that may occur in reality.

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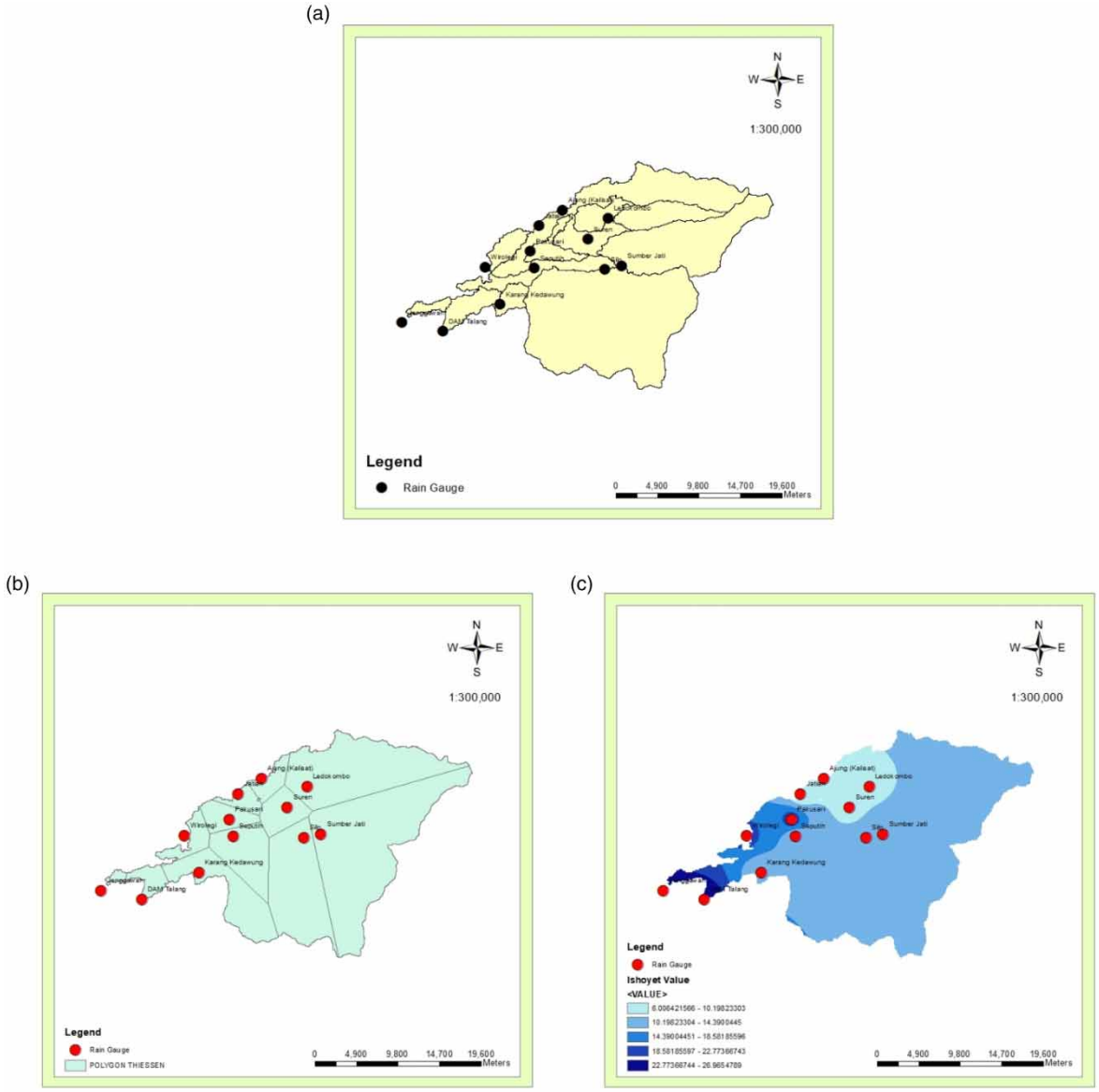


Figure 3 | Arithmetic (a) Thiessen Polygon, (b) Isohyet, and (c) method.

On the other hand, the Thiessen Polygon method divides the study region into polygons. The Thiessen Polygon method divides the study area into 12 different areas, each area represented by a single rain gauge, as shown in Figure 3(b). This rain gauge collects rainfall for the entire area contained within its polygon. The rain gauge rainfall value is then considered to represent the entire polygon uniformly. This approach takes rain gauge proximity into account and assumes that rainfall is consistent within each polygon. Every point in each polygon is closer to its enclosing sample point than to any other sampling point.

Meanwhile, by using the Isohyet method, the area is separated into subregions based on reported rainfall values, which are classified as very high, high, moderate, low, and very low, as shown in Figure 3(c). These categories are depicted by Isohyet contour lines. It is believed that each subregion within the Isohyet range will have identical rainfall characteristics. This method considers the spatial variability of rainfall. By using multiple rain gauges, it accounts for variations in rainfall across different locations.

3.2. Correlation of observational rainfall and GPM-IMERG satellite precipitation data

Correlation is used to determine the level of accuracy between satellite precipitation data and observational rainfall data. A good correlation value indicates better rainfall data. The result shows that daily correlations are generally low, indicating weak to moderate agreement between daily satellite data and ground observations, as shown in Figure 4. However, correlations improve significantly with 10-day aggregation and are highest for

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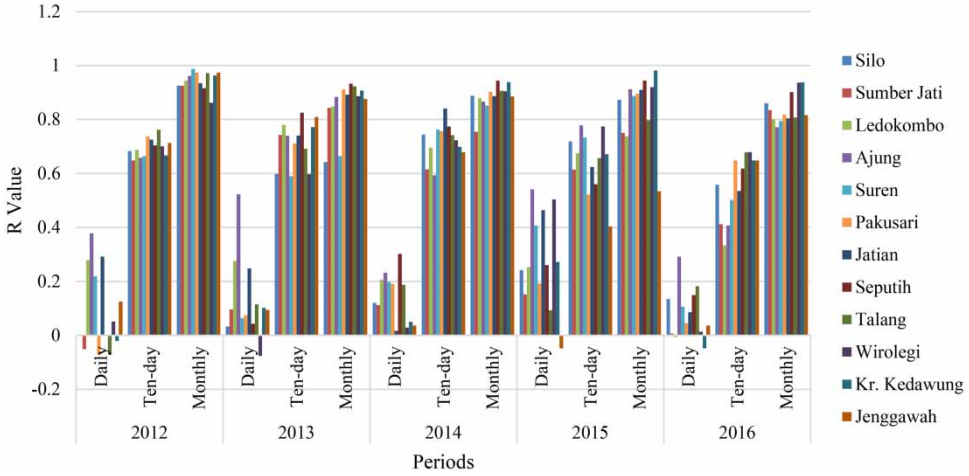


Figure 4 | Correlation result.

monthly data, showing strong to very strong correlations. The higher daily correlation value is below 0.5, which indicates a weak to moderate agreement between daily satellite data and ground observations. The 10-day correlation value has improved compared to daily, with many values exceeding 0.6. Furthermore, the monthly correlations value has the highest among the three, with most values above 0.8, which indicates strong to very strong correlations, reflecting better alignment over longer periods. This trend suggests that longer aggregation periods reduce short-term variability and provide a better match between satellite and observational data. Stations such as Ajung, Ledokombo, and Seputih show consistently strong correlations for 10-day and monthly periods, indicating reliable satellite data for these locations. In contrast, stations such as Silo, Sumber Jati, and Pakusari show significant improvement only with data aggregation, highlighting the need for careful calibration and validation for daily data.

3.3. Calibration

Calibration is used to improve simulation results to align them more precisely with observed values. The calibration parameters used in this study include max storage, CN, time of concentration (T_c), and storage coefficient. The calibration was conducted from 1 January 2012, to 31 December 2016, by considering outflow discharge in Talang station that was generated by daily rainfall. The rain gauge being analyzed is located at Seputih Station, which represents the upstream sub-watershed area that is indicated to have precipitation data (using satellite), but it is not recorded in the field. The parameter values obtained from the calculations were adjusted either by decreasing or increasing them as needed.

The NSE values for observational rainfall data using the Arithmetic, Thiessen Polygon, and Isohyet methods are 0.254, 0.439, and 0.212. However, with precipitation GPM-IMERG satellite data using these three methods, the NSE values were generally better than those for observational data, as shown in Table 4. Specifically, the Thiessen Polygon method showed the highest accuracy for both observational and GPM-IMERG data among the three mean areal precipitation methods. Furthermore, this study also evaluates the TRMM satellite for comparison. The Thiessen Polygon method with GPM-IMERG and TRMM satellite data resulted in NSE values of 0.515 and 0.445, respectively, indicating that GPM-IMERG data performed better. Additionally, observational rainfall data generally showed lower and negative percent bias values compared to GPM-IMERG and TRMM satellite data. This indicates that discharge responses from observational rainfall data tend to underestimate, while satellite precipitation data tend to overestimate, except for the Arithmetic and Isohyet methods.

The calibration results are shown by using daily rainfall data observed from Seputih station, as shown in Figures 5–8. The yellow line represents the daily rainfall or precipitation data at Seputih station, the black line represents the observed daily discharge at the Talang outlet, the green line represents the observational daily discharge, while the blue line represents the daily discharge generated using GPM-IMERG data, and the orange line represents the daily discharge generated using TRMM data. It is indicated that all model calibration results display a hydrological response pattern similar to the observed discharge. A significant difference is observed when

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Table 4 | Performance of model calibration

No	Input		RMSE	NSE	% Bias
	Method	**Data**			
1	Arithmetic	Observation	0.9	0.225	−7.915
		GPM-IMERG	1.2	0.254	−0.44
2	Thiessen Polygon	Observation	0.9	0.439	−7.46
		GPM-IMERG	0.6	0.515	6.13
3	Isohyet	Observation	1.2	0.111	−8.28
		GPM-IMERG	0.9	0.236	−5.70
4	Thiessen Polygon	TRMM	0.9	0.445	10.01
5	Thiessen Polygon (high flow)	Observation	0.9	0,270	−5.55
	Thiessen Polygon (low flow)		0.8	0,305	−6.41
6	Thiessen Polygon (high flow)	GPM-IMERG	0.7	0.451	−16.87
	Thiessen Polygon (low flow)		0.8	0.337	16.26

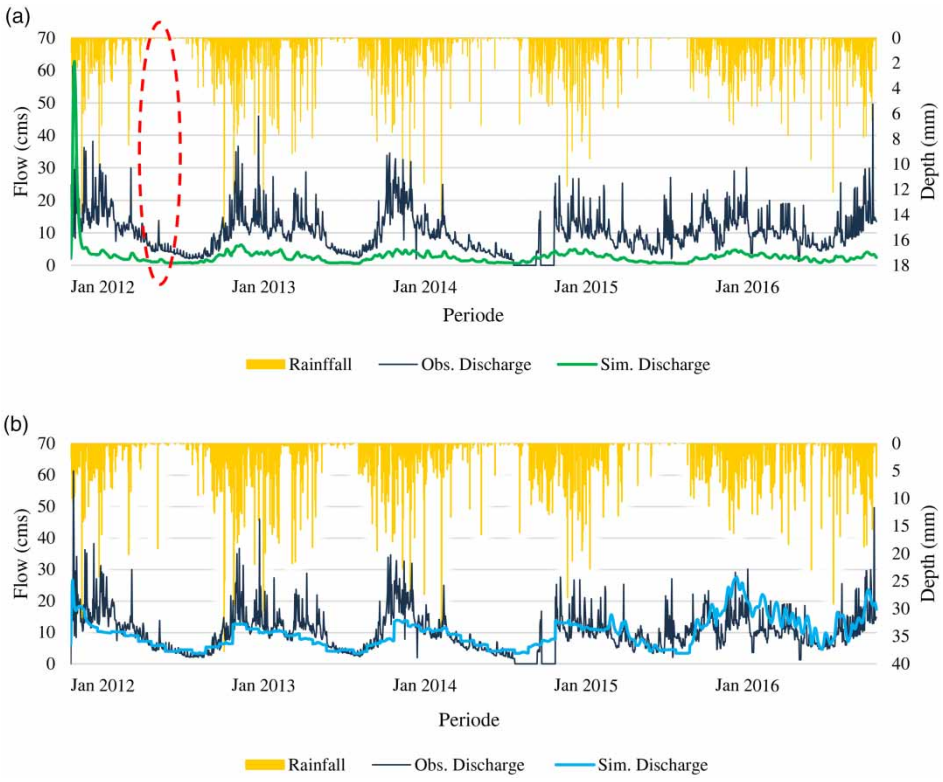


Figure 5 | Arithmetic (a) observation, and (b) GPM-IMERG results.

using the Thiessen Polygon method with GPM-IMERG input data, as shown in Figure 6(b), which provides good results and effectively represents extreme rainfall conditions. The poor performance of observational data as input for the HEC-HMS model is due to the lack of observed rainfall data, particularly around July 2012, as marked by red circles as shown in Figures 5(a), 6(a), and 7(a). In contrast, on the same dates, GPM-IMERG and TRMM rainfall data successfully detected rainfall and observed discharge, as shown in Figures 5(b), 6(b), 7(b), and 8.

Based on the previous study by using other satellite data such as TRMM in Bedadung watershed show that hydrological modeling performance using manual rainfall recorder (MRR) rain data input was better than the performance using satellite data TRMM 3B42 (Alhamda *et al.* 2020). The R^2 and NSE values for MRR rain data input were 0.72 and 0.71. In contrast, the R^2 and NSE values for the TRMM B342 satellite rain data

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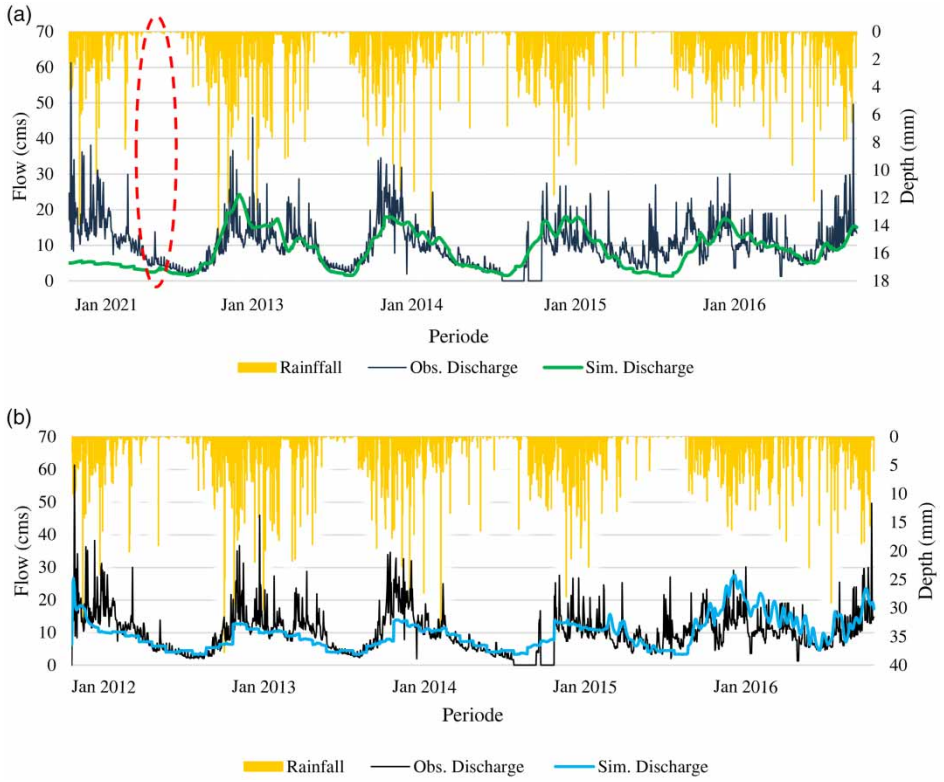


Figure 6 | Thiessen Polygon (a) observation, and (b) GPM-IMERG results.

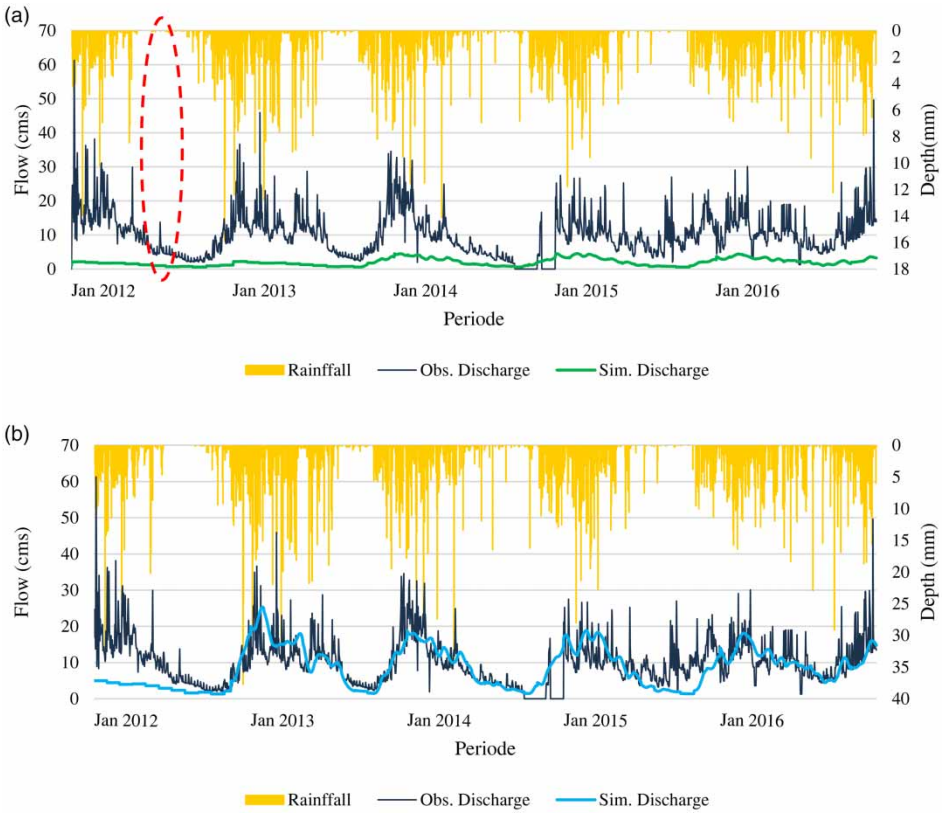


Figure 7 | Isohyet (a) observation; and (b) GPM-IMERG results.

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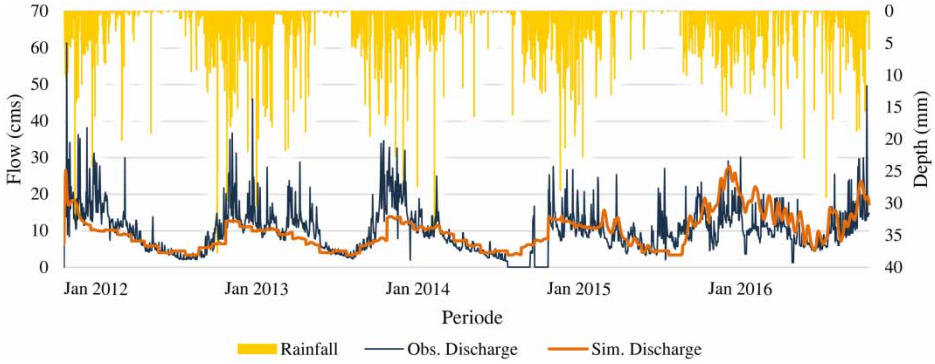


Figure 8 | Thiessen Polygon TRMM results.

input were 0.57 and 0.55, respectively. In other studies in the Sampean Baru, Bedadung, and Mayang watersheds, the MMR provided a better discharge response in all watersheds compared to TRMM and GPM, however, in terms of satellite products, the GPM-3IMERGDF showed slightly better results than the TRMM 3B42 (Hidayah *et al.* 2021).

Therefore, by using the Thiessen Polygon method, the consistency of GPM-IMERG precipitation data in representing locations without rain gauges is tested by comparing it with observation data based on the average rainfall over the Thiessen Polygon for daily data during the rainy season from October 2012 to March 2013 and the dry season from April to September 2013. The results show that the observed discharge and the model exhibit a similar pattern, as shown in Figures 9 and 10. It shows that the GPM-IMERG precipitation data is more accurate during the rainy season compared to the observation data, whereas the observation data is more accurate during the dry season.

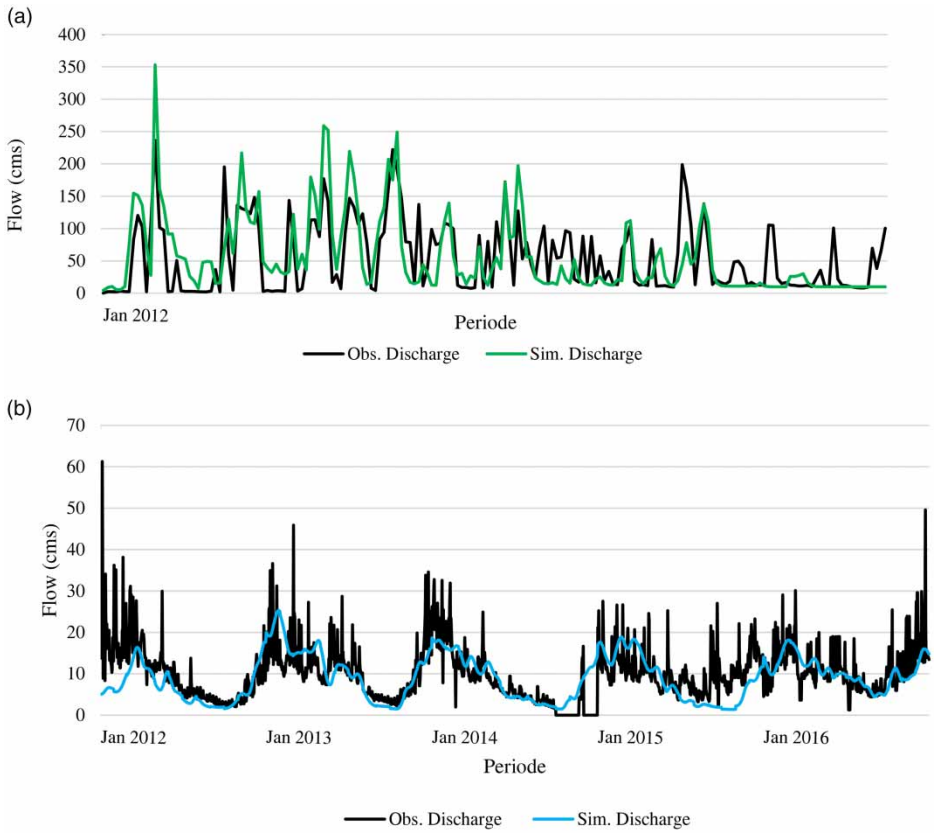


Figure 9 | Thiessen Polygon high flow (a) observation; and (b) GPM-IMERG.

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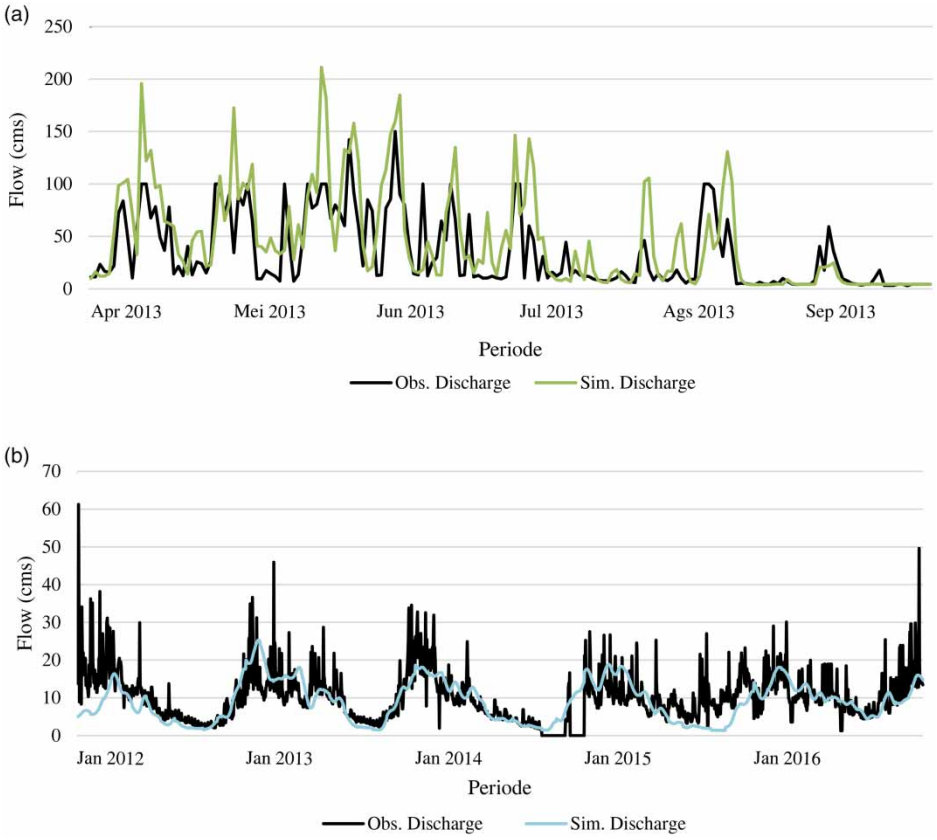


Figure 10 | Thiessen Polygon low flow (a) observation; and (b) GPM-IMERG.

3.4. Validation

The validation process took place between 1 January 2017, and 31 December 2021. It involved using daily input data and calibrated parameter values collected earlier. The validation results indicate that the NSE value for observational rainfall data using the Thiessen Polygon method is 0.481. For GPM-IMERG precipitation data, the NSE values using the Arithmetic, Isohyet, and Thiessen Polygon methods are 0.392, 0.333, and 0.655, respectively. The Thiessen Polygon method using GPM-IMERG data shows a higher NSE value than the observational data, as shown in Table 5. Additionally, the validation results indicate that all treatments show an overestimation, except for GPM-IMERG using the Isohyet method, as shown in Figure 11.

Table 5 | Model validation performance

No	Input	RMSE	NSE	% Bias
1	Arithmetic GPM-IMERG	1	0.392	35.95
2	Thiessen Polygon Observation	0.8	0.481	19.98
	Thiessen Polygon GPM-IMERG	0.8	0.655	21.55
3	Isohyet GPM-IMERG	0.8	0.333	−22.53

In the Arithmetic method, the GPM-IMERG precipitation data in the upstream area provides a more accurate representation of rainfall events compared to the average observed rainfall, as it disregards the influence of area. The absence of rain gauges in the upstream region led to the data acquired by the satellite not being recorded in the gauges, as shown in Figures 5 and 11(a). However, the impact of dividing the area into polygons using the Thiessen Polygon method primarily reflects the occurrence of rainfall events, rather than the absence of observed rainfall data in the upstream area, as shown in Figures 6 and 11(b). Given that each polygon is associated with a representative rain gauge, it is possible to retrieve the missing data from the upstream area. This data indicates a

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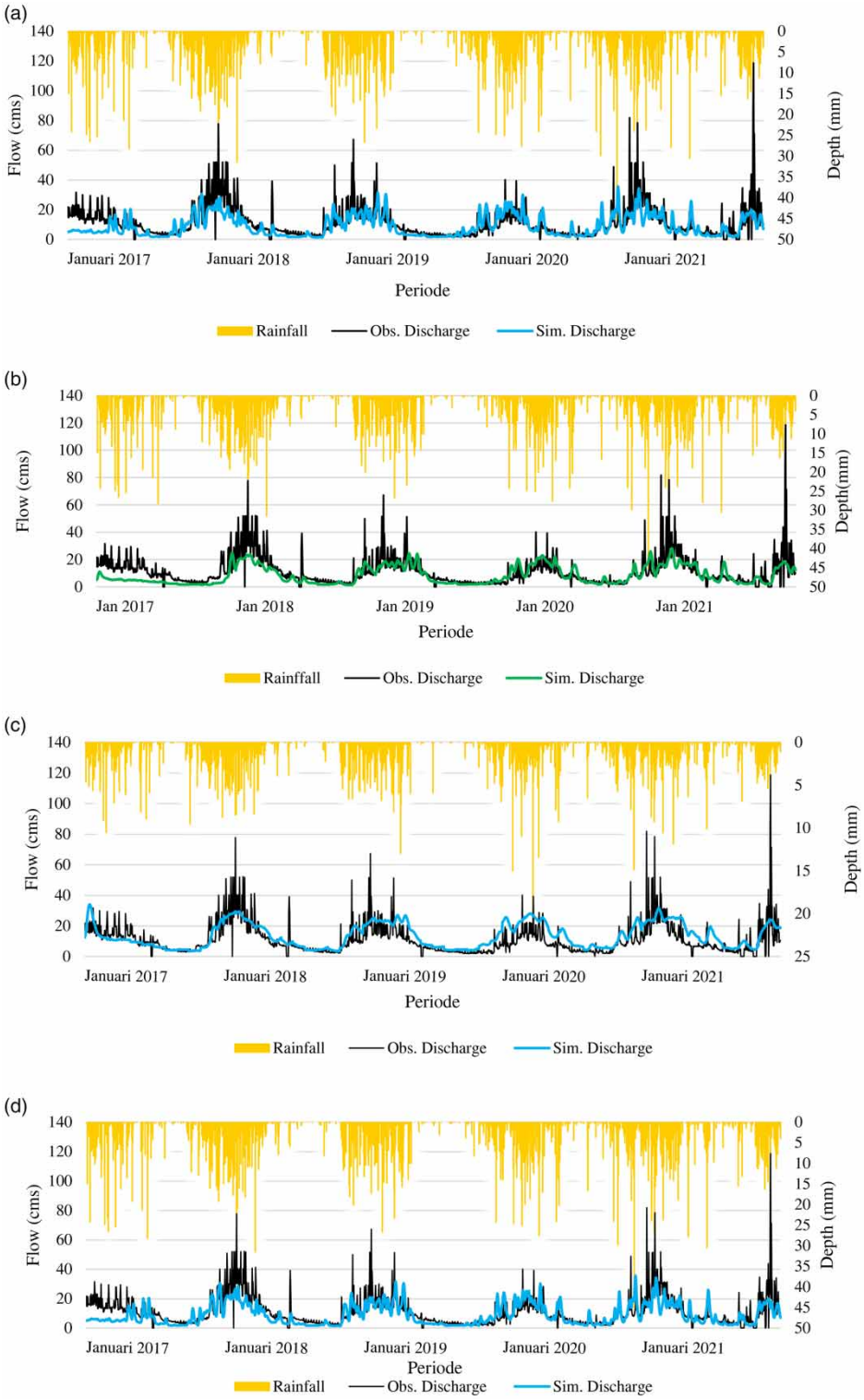


Figure 11 | (a) Arithmetic GPM-IMERG, (b) Thiessen Polygon Observation, (c) Thiessen Polygon GPM-IMERG, and (d) Isohyet GPM-IMERG validation results.

greater NSE value compared to the Arithmetic and Isohyet techniques. The Isohyet approach fails to adequately depict the upstream rainfall value, which has a significant impact region, using only four rainfall stations. The upstream region widens as it is divided by the Isohyet line. Consequently, the increased size of the area results in a greater amount of unquantified precipitation in the upstream regions, thereby reducing its significance compared to the Thiessen Polygon technique."

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3.5. Sensitivity analysis result

Sensitivity analysis was conducted on simulation scenarios using GPM-IMRG input data, focusing on volume discharge and peak discharge as objective functions. The slope of the tangent lines on the curve indicates that the steeper the slope of a parameter's tangent line, the greater its sensitivity, as shown in Figure 12. The increase in parameter values during calibration is reflected in the Nash and Sutcliffe values. The sensitivity ranking of parameters affecting volume discharge in the Mayang watershed, from highest to lowest, is SCS–CN > SCS UH–LT > SCS CN–Ia > SC–MS > SC–ISD > SS–MS > SC–CC > SS–SD. For peak discharge, the order is SCS–CN > SCS UH–LT > SCS CN–Ia > SC–MS > SC–ISD > SC–CC > SS–MS > SS–ISD. In both cases, the most influential parameters are SCS–CN and SCS UH–LT. However, there is a slight difference in the influence of SCS CN–Ia (9.98% higher) and SC–MS (10.68% higher) on volume discharge compared to peak discharge. Comparing the sensitivity of these parameters, it is evident that SCS CN and SCS UH-LT are the primary factors affecting the model, indicating that volume runoff and direct runoff have a greater impact on peak discharge.

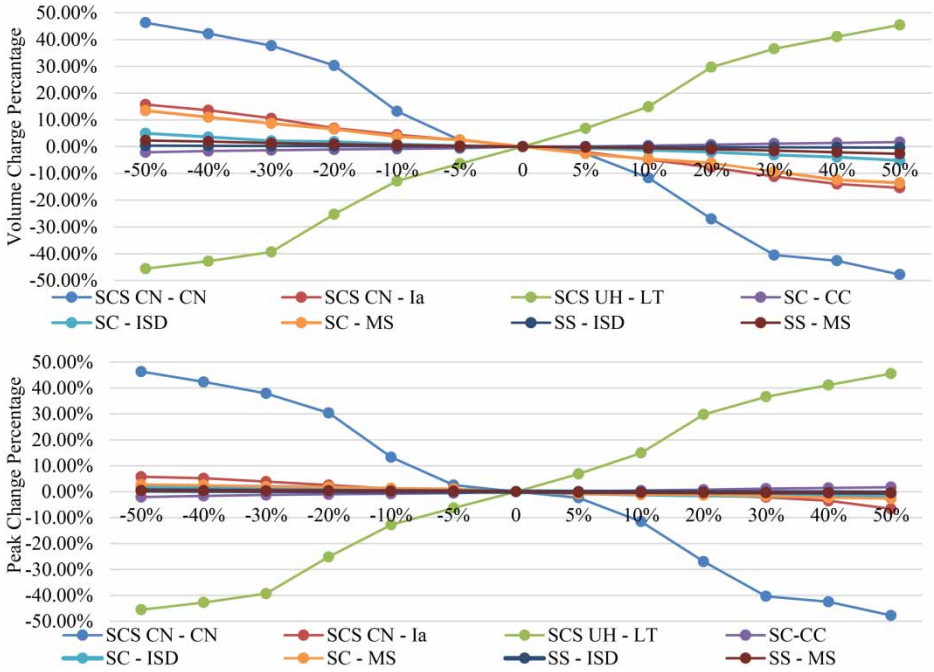


Figure 12 | Sensitivity analysis for GPM-IMERG precipitation data.

4. DISCUSSION

Spatial modeling of hydrology shows the importance of the availability of rain gauges that can represent the entire sub-watershed, which can be demonstrated by the Thiessen Polygon result. The rainfall data used for this runoff model simulation has been tested for correlation at all rainfall stations, resulting in a good match between satellite data and observational data. These patterns determine how rainfall translates into runoff and other hydrological processes. Furthermore, based on Michelon *et al.* (2021), high-density rainfall observations are crucial for accurately capturing these patterns, which are particularly challenging to observe in mountainous environments due to the complex terrain and variability in rainfall distribution. When the density of rain gauges is low or when gauges are irregularly distributed, the precipitation data collected is less representative of the actual spatial variability of rainfall, leading to significant uncertainties in the hydrological model parameters (Zeng *et al.* 2018). Inaccurate precipitation inputs cause the model to calibrate parameters incorrectly, resulting in poor simulation of hydrological processes like runoff. Thus, observing high-density rainfall, including the distance of rainfall from the outlet, is key to predicting hydrological responses in terms of runoff coefficients and lag time in small mountainous catchment areas. Increasing the density of stations will improve the performance of hydrological modeling; however, it is important to remember that adequate station density depends on the catchment area, type of event, and distribution of stations (Hohmann *et al.* 2021). Recent studies highlight the variability in the

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optimum number of rain gauges, which ranged from 14 km^2/gauge to 38 km^2/gauge, with an average of 25 km^2/gauge. It was found that the current rain gauge density needs to be improved by at least a factor of three to mimic the full-scale level (Gyasi-Agyei 2020).

The research on rainfall modeling in streams using GPM-IMERG precipitation data input with the HEC-HMS program yielded satisfactory modeling performance when compared to observed rainfall input. The simulated discharge pattern, using observational rainfall data and GPM-IMERG satellite precipitation data as input, is compatible with the observed discharge, as indicated by the calibration and validation findings. Under conditions of low flow, the simulated release values align with the observed discharge when using both the observed rainfall data and the GPM-IMERG satellite precipitation data. During the highest flow rate, the simulated discharge value does not match the observed discharge when using observational rainfall data as input. However, when using GPM-IMERG satellite precipitation data as input, the simulated discharge aligns well with the observed discharge. Furthermore, the temporal influence shows that high-flow conditions better represent the hydrological response compared to low flow conditions. This result aligns with a study by Li *et al.* (2021), which GPM-IMERG data was effective in simulating hydrological responses during high-flow conditions and showed rational flow simulations with notable seasonal and regional variations in performance in Taiwan. The study in hydrological modeling over the Nanliu River Basin that evaluates the performance of the GPM-IMERG products against gauge observations found that when driven by IMERG data, the hydrological model's performance improved for high-flow simulations compared to low flow simulations (Tong *et al.* 2018). However, the temporal and spatial resolution of satellite data, while suitable for capturing high-flow events that involve significant precipitation over large areas, may not be sufficient for accurately representing the finer details of low flow conditions. Smaller, localized precipitation events can be averaged out or missed in the data processing, leading to inaccuracies (Ahmed *et al.* 2022). Low-intensity precipitation events, which are characteristic of low flow conditions, can be challenging to detect accurately due to the limited sensitivity of the satellite sensors, which can lead to underestimation or missing detection of light rain events, impacting the overall accuracy of low flow simulations (Zhou *et al.* 2021).

Despite limitations in detecting low flow conditions, GPM-IMERG outperforms the TRMM satellite in producing more accurate precipitation estimates. In this study, the RMSE, NSE, and %Bias of Thiessen Polygon GPM-IMERG are 0.6, 0.515, and 6.13, respectively, while for Thiessen Polygon TRMM, these values are 0.9, 0.445, and 10.01. Thus, GPM-IMERG demonstrates better performance compared to TRMM, indicating more accurate, reliable, and less biased precipitation estimates. In previous studies, such as in the Bedadung Watershed, Jember Regency, Indonesia, GPM-IMERG also showed a better fit than TRMM for various hydrological models such as Variable Infiltration Capacity (VIC), HEC-HMS, and SWAT, suggesting that it provides a more accurate representation of rainfall data in certain contexts (Hidayah *et al.* 2021). According to Gaona *et al.* (2016), GPM offers more comprehensive global coverage than TRMM, allowing GPM-IMERG to capture precipitation data over a larger portion of the Earth's surface, including higher latitude regions that experience significant precipitation events. Additionally, GPM-IMERG enhances the spatial resolution of precipitation estimates to $0.1° \times 0.1°$ and provides data at finer temporal resolutions, such as every 30 min. This is a significant improvement over TRMM, which had a spatial resolution of 0.25° and temporal resolution of 3 h, allowing for more detailed and accurate precipitation measurements (Chen & Li 2016). In comparison to other satellites such as PERSIANN, GPM-IMERG offers more frequent updates and higher resolution data (Pradhan *et al.* 2021). While GPM-IMERG provides high-resolution and frequent data, it may still miss localized precipitation events or have inaccuracies due to the complexity of weather systems, such as during the dry season and due to topographical influences – factors that can similarly affect other satellite-based estimates such as TRMM and PERSIANN (Zhou *et al.* 2023). Therefore, further studies should compare satellite rainfall data with other emerging rainfall data products such as microwave or radar rainfall, Climate Hazards Group Infrared Precipitation with Stations, CHAOS simulation (CHAOS-hydro), and the XPOL weather radar (XPOL-hydro) to evaluate their accuracy and reliability in hydrological modeling (Thorndahl *et al.* 2017; Varlas *et al.* 2019).

Based on the sensitivity results, the CN is a crucial factor as it measures the runoff potential in a region. Similar to findings from the Meenachil River basin study, CN and lag time are the most sensitive parameters in the HEC-HMS model for predicting peak discharge volume (George *et al.* 2022). A higher CN value implies a greater tendency for runoff (Kristanto *et al.* 2021), resulting in less water infiltrating the soil and more water flowing over the surface, leading to a higher volume of river discharge (Priambodo *et al.* 2021). The increase in runoff and discharge is often exacerbated by the lack of infiltration zones, such as permeable soils or vegetation, which

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would otherwise absorb some of the rainfall. Additionally, the time of concentration (T_c) and the storage coefficient (R) are crucial variables impacting the timing of the simulated peak discharge. T_c represents the time required for water to travel from the farthest point in the watershed to the outflow, while the storage coefficient indicates the rate at which storage areas release water. Increasing T_c and R values extends the time to reach peak discharge, affecting the model's ability to accurately predict flood peaks and the timing of runoff events. During the calibration phase, simulated discharge levels are compared to observed data, including values from observational precipitation data, satellite precipitation data, and recorded discharge measurements (Esmali *et al.* 2021). Even so, satellite precipitation data, especially GPM-IMERG, can be effectively used for hydrological modeling in regions with sparse observational data. Overestimated precipitation data using the Thiessen Polygon method is beneficial for flood management planning, as it ensures preparedness for potential extreme weather events. On the other hand, underestimated precipitation data using the Isohyet method is useful for water resource planning, including irrigation, raw water supply, and hydropower plant management, as it encourages efficient and sustainable water use.

5. CONCLUSIONS

GPM-IMERG satellite precipitation data proves to be a valuable tool for hydrological modeling, especially in areas with sparse observational data. Among the methods evaluated, the Thiessen Polygon method was particularly effective when paired with GPM-IMERG satellite data, providing more reliable precipitation data by accounting for the proximity of rain gauges. Calibration and validation results further confirmed the superiority of GPM-IMERG data over observational data, especially when using the Thiessen Polygon method, enhancing the accuracy of simulated discharge patterns, particularly under high-flow conditions. Despite some limitations in capturing low-intensity precipitation events, GPM-IMERG outperforms other satellite products like TRMM in terms of accuracy. Sensitivity analysis revealed that parameters such as CN and time of concentration significantly influence model outcomes. The integration of GPM-IMERG data in hydrological modeling offers valuable insights for flood management and water resource planning, particularly in regions with sparse observational data, ensuring better preparedness for extreme weather events and promoting sustainable water use practices. Further studies should explore emerging rainfall data products, such as microwave or radar rainfall, to continue improving hydrological modeling accuracy. The next development of this research is to perform more advanced rainfall prediction modeling by leveraging various features from satellite data and other meteorological data using machine learning techniques such as deep neural networks or least mean square training.

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DATA AVAILABILITY STATEMENT

Data cannot be made publicly available; readers should contact the corresponding author for details.

CONFLICT OF INTEREST

The authors declare there is no conflict.

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